Vector DBs: Podcast Recommender System — Queries and Results

PostgreSQL 17 + pgvector, L2 distance (<->) for every query. 346 podcasts, 832,839 segments. The query segment is excluded from Q1–Q4; the query segment's or episode's own podcast is excluded from Q5–Q6. Times are in seconds; floats rounded for display.

Q1) Five most similar segments to segment "267:476"

Input: "that if we were to meet alien life at some point"

Query (`SIMILAR_SEGMENTS` in db_query.py):

```
SELECT p.title,
        s.id,
        s.content,
        s.start_time,
        s.end_time,
        s.embedding <-> (SELECT embedding FROM segment WHERE id = %(seg_id)s) AS distance
    FROM segment s
    JOIN podcast p ON s.podcast_id = p.id
    WHERE s.id != %(seg_id)s
    ORDER BY distance
    LIMIT 5;
```

Results for seg_id = '267:476':

title	id	content	start_time	end_time	distance
Ryan Graves: UFOs, Fighter Jets, and Aliens | Lex Fridman Podcast #308	113:2792	encounters, human beings, if we were to meet another alien	6725.62	6729.86	0.6483
Richard Dawkins: Evolution, Intelligence, Simulation, and Memes | Lex Fridman Podcast #87	268:1019	Suppose we did meet an alien from outer space	2900.04	2903.08	0.6558
Jeffrey Shainline: Neuromorphic Computing and Optoelectronic Intelligence | Lex Fridman Podcast #225	305:3600	but if we think of alien civilizations out there	9479.96	9484.04	0.6595
Michio Kaku: Future of Humans, Aliens, Space Travel & Physics | Lex Fridman Podcast #45	18:464	So I think when we meet alien life from outer space,	1316.86	1319.58	0.6662
Alien Debate: Sara Walker and Lee Cronin | Lex Fridman Podcast #279	71:989	because if aliens come to us	2342.34	2343.62	0.6743

Q2) Five most dissimilar segments to segment "267:476"

Input: "that if we were to meet alien life at some point"

Query (`DISSIMILAR_SEGMENTS` in db_query.py):

```
SELECT p.title,
        s.id,
        s.content,
        s.start_time,
        s.end_time,
        s.embedding <-> (SELECT embedding FROM segment WHERE id = %(seg_id)s) AS distance
    FROM segment s
    JOIN podcast p ON s.podcast_id = p.id
    WHERE s.id != %(seg_id)s
    ORDER BY distance DESC
    LIMIT 5;
```

Results for seg_id = '267:476':

title	id	content	start_time	end_time	distance
Jason Calacanis: Startups, Angel Investing, Capitalism, and Friendship | Lex Fridman Podcast #161	119:218	a 73 Mustang Grande in gold?	519.96	523.80	1.6158
Rana el Kaliouby: Emotion AI, Social Robots, and Self-Driving Cars | Lex Fridman Podcast #322	133:2006	for 94 car models.	5818.62	5820.82	1.5863
Travis Stevens: Judo, Olympics, and Mental Toughness | Lex Fridman Podcast #223	283:1488	when I called down to get the sauna.	3709.34	3711.10	1.5726
Jeremy Howard: fast.ai Deep Learning Courses and Research | Lex Fridman Podcast #35	241:1436	which has all the courses pre-installed.	4068.90	4071.14	1.5663
Joscha Bach: Nature of Reality, Dreams, and Consciousness | Lex Fridman Podcast #212	307:3933	and very few are first class and some are budget.	10648.64	10650.96	1.5616

Q3) Five most similar segments to segment "48:511"

Input: "Is it is there something especially interesting and profound to you in terms of our current deep learning neural network, artificial neural network approaches and the whatever we do understand about the biological neural network."

Query (`SIMILAR_SEGMENTS` in db_query.py):

```
SELECT p.title,
        s.id,
        s.content,
        s.start_time,
        s.end_time,
        s.embedding <-> (SELECT embedding FROM segment WHERE id = %(seg_id)s) AS distance
    FROM segment s
    JOIN podcast p ON s.podcast_id = p.id
    WHERE s.id != %(seg_id)s
    ORDER BY distance
    LIMIT 5;
```

Results for seg_id = '48:511':

title	id	content	start_time	end_time	distance
Andrew Huberman: Neuroscience of Optimal Performance | Lex Fridman Podcast #139	155:648	Is there something interesting to you or fundamental to you about the circuitry of the brain	3798.48	3805.84	0.6523
Cal Newport: Deep Work, Focus, Productivity, Email, and Social Media | Lex Fridman Podcast #166	61:3707	of what we might discover about neural networks?	8498.02	8500.10	0.7121
Matt Botvinick: Neuroscience, Psychology, and AI at DeepMind | Lex Fridman Podcast #106	48:512	And our brain is there. There's some there's quite a few differences. Are some of them to you either interesting or perhaps profound in terms of in terms of the gap we might want to try to close in trying to create a human level intelligence.	1846.84	1865.84	0.7196
Yann LeCun: Dark Matter of Intelligence and Self-Supervised Learning | Lex Fridman Podcast #258	276:2642	Have these, I mean, small pockets of beautiful complexity. Does that, do cellular automata, do these kinds of emergence and complex systems give you some intuition or guide your understanding of machine learning systems and neural networks and so on?	8628.16	8646.16	0.7357
Stephen Wolfram: Fundamental Theory of Physics, Life, and the Universe | Lex Fridman Podcast #124	2:152	So is there something like that with physics where so deep learning neural networks have been around for a long time?	610.86	618.86	0.7367

Q4) Five most similar segments to segment "51:56"

Input: "But what about like the fundamental physics of dark energy? Is there any understanding of what the heck it is?"

Query (`SIMILAR_SEGMENTS` in db_query.py):

```
SELECT p.title,
        s.id,
        s.content,
        s.start_time,
        s.end_time,
        s.embedding <-> (SELECT embedding FROM segment WHERE id = %(seg_id)s) AS distance
    FROM segment s
    JOIN podcast p ON s.podcast_id = p.id
    WHERE s.id != %(seg_id)s
    ORDER BY distance
    LIMIT 5;
```

Results for seg_id = '51:56':

title	id	content	start_time	end_time	distance
George Hotz: Hacking the Simulation & Learning to Drive with Neural Nets | Lex Fridman Podcast #132	308:144	I mean, we don't understand dark energy, right?	500.44	502.60	0.6682
Lex Fridman: Ask Me Anything - AMA January 2021 | Lex Fridman Podcast	243:273	Like, what's up with this dark matter and dark energy stuff?	946.22	950.12	0.7356
Katherine de Kleer: Planets, Moons, Asteroids & Life in Our Solar System | Lex Fridman Podcast #184	196:685	being like, what the hell is dark matter and dark energy?	2591.72	2595.96	0.7631
Alex Filippenko: Supernovae, Dark Energy, Aliens & the Expanding Universe | Lex Fridman Podcast #137	51:36	Do we have any understanding of what the heck that thing is?	216.00	219.00	0.7922
Leonard Susskind: Quantum Mechanics, String Theory and Black Holes | Lex Fridman Podcast #41	122:831	That is a big question in physics right now.	2374.90	2377.62	0.8023

Q5) Five most similar podcast episodes to each segment (episode embeddings averaged)

Query (`SIMILAR_EPISODES_TO_SEGMENT` in db_query.py):

```
SELECT p.title,
       AVG(s.embedding) <-> (SELECT embedding FROM segment WHERE id = %(seg_id)s) AS distance
    FROM segment s
    JOIN podcast p ON s.podcast_id = p.id
    WHERE s.podcast_id != (SELECT podcast_id FROM segment WHERE id = %(seg_id)s)
    GROUP BY p.id, p.title
    ORDER BY distance
    LIMIT 5;
```

Results for seg_id = '267:476' — "that if we were to meet alien life at some point":

title	distance
Sara Walker: The Origin of Life on Earth and Alien Worlds | Lex Fridman Podcast #198	0.7829
Martin Rees: Black Holes, Alien Life, Dark Matter, and the Big Bang | Lex Fridman Podcast #305	0.7879
Max Tegmark: Life 3.0 | Lex Fridman Podcast #1	0.7887
Sean Carroll: The Nature of the Universe, Life, and Intelligence | Lex Fridman Podcast #26	0.7891
Nick Bostrom: Simulation and Superintelligence | Lex Fridman Podcast #83	0.7911

Results for seg_id = '48:511' — "Is it is there something especially interesting and profound to you in...":

title	distance
Christof Koch: Consciousness | Lex Fridman Podcast #2	0.7538
Dileep George: Brain-Inspired AI | Lex Fridman Podcast #115	0.7605
Tomaso Poggio: Brains, Minds, and Machines | Lex Fridman Podcast #13	0.7616
Elon Musk: Neuralink, AI, Autopilot, and the Pale Blue Dot | Lex Fridman Podcast #49	0.7762
Philip Goff: Consciousness, Panpsychism, and the Philosophy of Mind | Lex Fridman Podcast #261	0.7872

Results for seg_id = '51:56' — "But what about like the fundamental physics of dark energy? Is there a...":

title	distance
Sean Carroll: Quantum Mechanics and the Many-Worlds Interpretation | Lex Fridman Podcast #47	0.7767
Stephen Wolfram: Fundamental Theory of Physics, Life, and the Universe | Lex Fridman Podcast #124	0.8081
Donald Hoffman: Reality is an Illusion - How Evolution Hid the Truth | Lex Fridman Podcast #293	0.8166
Cumrun Vafa: String Theory | Lex Fridman Podcast #204	0.8173
Avi Loeb: Aliens, Black Holes, and the Mystery of the Oumuamua | Lex Fridman Podcast #154	0.8255

Q6) Five most similar podcast episodes to episode VeH7qKZr0WI

Input: "Balaji Srinivasan: How to Fix Government, Twitter, Science, and the FDA | Lex Fridman Podcast #331"

Query (`SIMILAR_EPISODES_TO_EPISODE` in db_query.py):

```
SELECT p.title,
        AVG(s.embedding) <-> (SELECT AVG(embedding) FROM segment WHERE podcast_id = %(podcast_id)s) AS distance
    FROM segment s
    JOIN podcast p ON s.podcast_id = p.id
    WHERE s.podcast_id != %(podcast_id)s
    GROUP BY p.id, p.title
    ORDER BY distance
    LIMIT 5;
```

Results for podcast_id = 'VeH7qKZr0WI':

title	distance
Tyler Cowen: Economic Growth & the Fight Against Conformity & Mediocrity | Lex Fridman Podcast #174	0.1195
Eric Weinstein: Difficult Conversations, Freedom of Speech, and Physics | Lex Fridman Podcast #163	0.1257
Michael Malice and Yaron Brook: Ayn Rand, Human Nature, and Anarchy | Lex Fridman Podcast #178	0.1284
Steve Keen: Marxism, Capitalism, and Economics | Lex Fridman Podcast #303	0.1292
Michael Malice: The White Pill, Freedom, Hope, and Happiness Amidst Chaos | Lex Fridman Podcast #150	0.1304